

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
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Section / Batch:	BTECH - IT	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

An e-commerce system stores order IDs unsorted due to continuous transactions and uses Linear Search for retrieval.

a) Explain how Linear Search operates in this case. b) Analyze performance degradation as data grows. c) Suggest improvements and justify.

a) Explain how Linear Search operates in this case

In an e-commerce system, order IDs are stored in an unsorted list because new orders are added continuously. Since the data is not arranged in any order, Linear Search checks each order ID one by one from the beginning until the required order ID is found or the list ends.

Steps

- Start from the first order ID.
- Compare it with the target order ID.
- If it matches, return the position of the order.
- If it does not match, move to the next order ID.
- Repeat until the order is found or all IDs are checked.

Example

Order IDs: 1025, 1018, 1042, 1031, 1050

Search for 1031

- Compare 1025 → Not found
- Compare 1018 → Not found
- Compare 1042 → Not found
- Compare 1031 → Found

The order is found after 4 comparisons.

b) Analyze performance degradation as data grows

As the number of orders increases, Linear Search becomes slower because it may need to compare every order ID before finding the required one.

- Best Case: $O(1)$ – Order found at the first position.
- Average Case: $O(n)$ – Order found in the middle.
- Worst Case: $O(n)$ – Order found at the last position or not present.

Performance Analysis

Number of Orders Maximum Comparisons

100	100
1,000	1,000
10,000	10,000
100,000	100,000

As the dataset grows, the search time increases linearly, making Linear Search inefficient for large databases.

c) Suggest improvements and justify

- Use Binary Search after sorting the order IDs. It reduces search time to $O(\log n)$.
- Use a Hash Table (HashMap) for direct access. The average search time is $O(1)$.
- Use Database Indexing to retrieve records quickly without scanning the entire database.

These methods improve search speed and make the system more efficient for handling large numbers of orders.

Time Complexity

Search Method Best Case Average Case Worst Case

Linear Search	$O(1)$	$O(n)$	$O(n)$
Binary Search	$O(1)$	$O(\log n)$	$O(\log n)$
Hash Table	$O(1)$	$O(1)$	$O(n)$ (rare)

Conclusion: Linear Search is suitable for small, unsorted datasets. For large e-commerce systems, Binary Search, Hash Tables, or Database Indexing provide much faster retrieval.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____